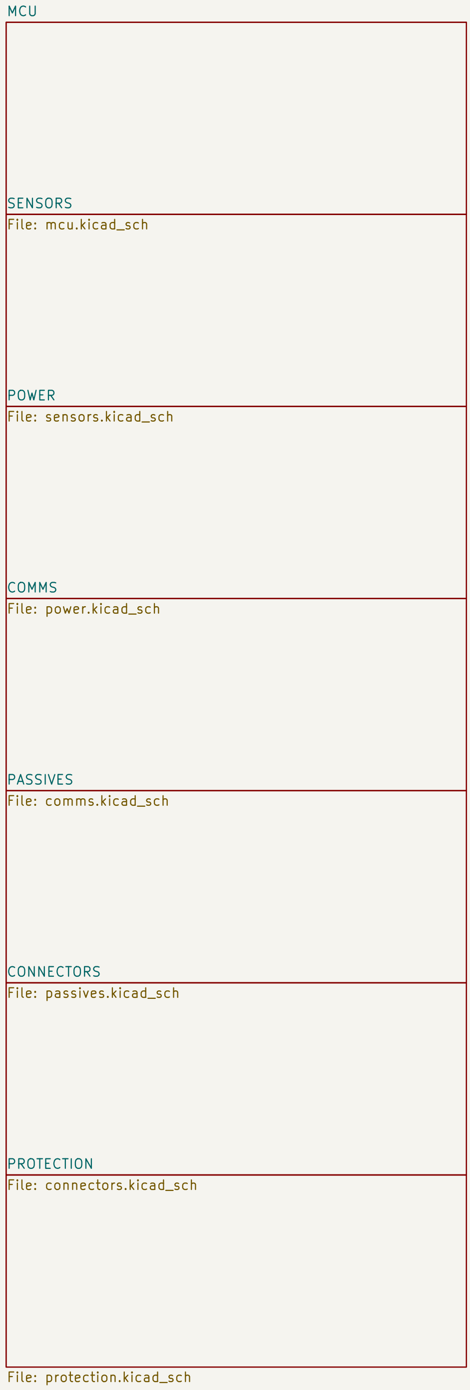
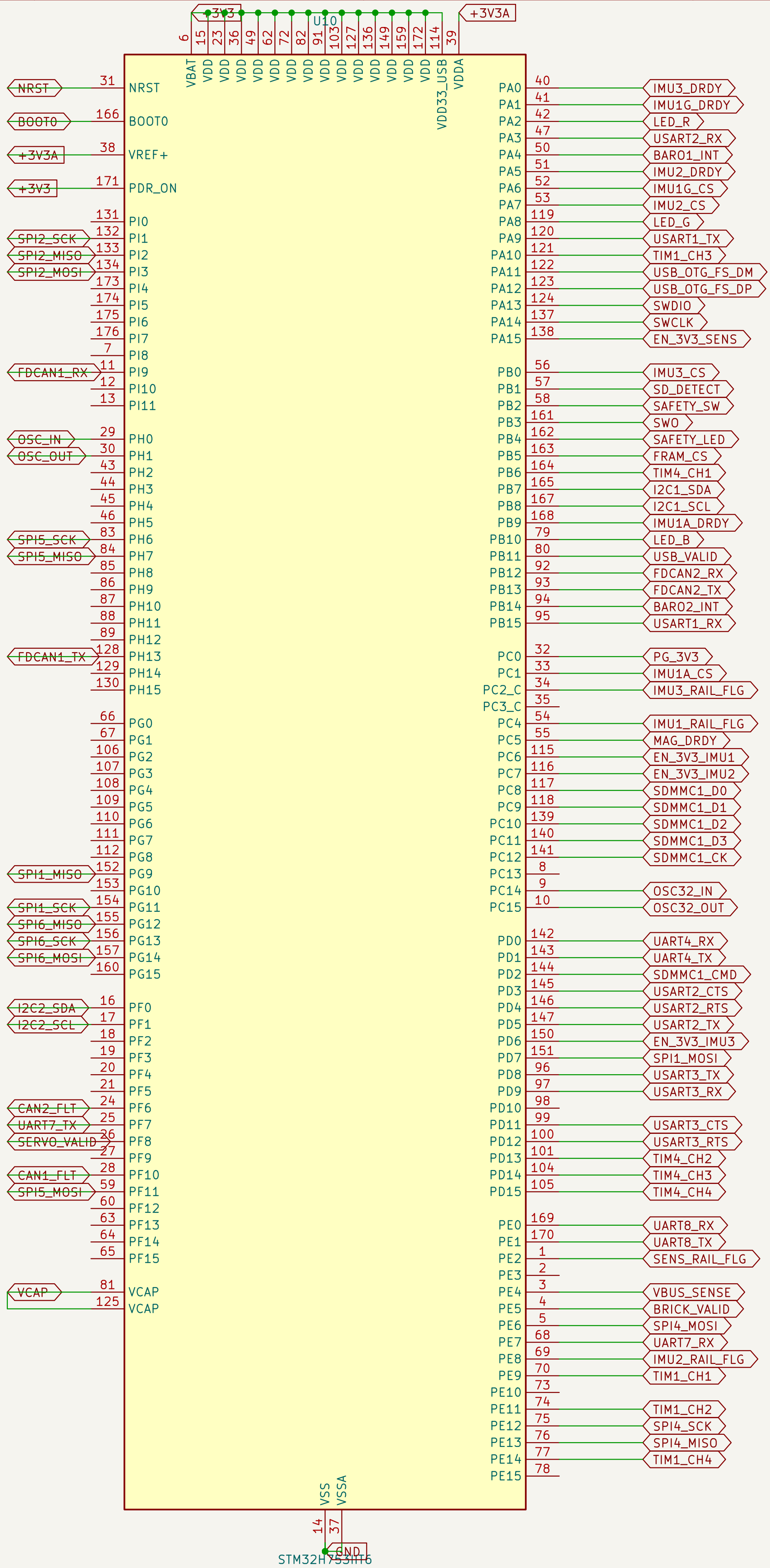
JFOX-FMU v1
STM32H753IIT6 flight controller - three IMUs, two barometers,
CAN FD with jumpered termination, 1 MB RAM.

Generated by hardware/tools/gen_fmu_schematic.py - do not hand-edit,
regenerate. Pin assignment comes from the part's own alternate-function
table (PINMAP.md); sensor pinouts from the datasheets (PINOUTS.md).

Cross-sheet nets are global labels, which is safe here because this is
one board - unlike the 3-board TMR project, where a global would short
all three instances together.



Sheet 1 of 8 Generated by hardware/tools/gen_fmu_schematic.py - regenerate, do not edit STM32H753IIT6 - 3 IMUs, 2 barometers, magnetometer, CAN FD **JFOX Aircraft Co., Ltd.**		
Sheet: / File: jfox-fmu.kicad_sch		
Title: JFOX-FMU v1		
Size: A4	Date: 2026-08-03	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 1/8



VCAP1/VCAP2 are the internal LDO output. NOT a supply – do not tie to +3V3
Pins allocated from the part's own AF table – see PINMAP.md
STM32H753IIT6 LQFP176 – 1 MB RAM, 2 MB flash, Cortex–M7 480 MHz

JFOX Aircraft Co., Ltd.

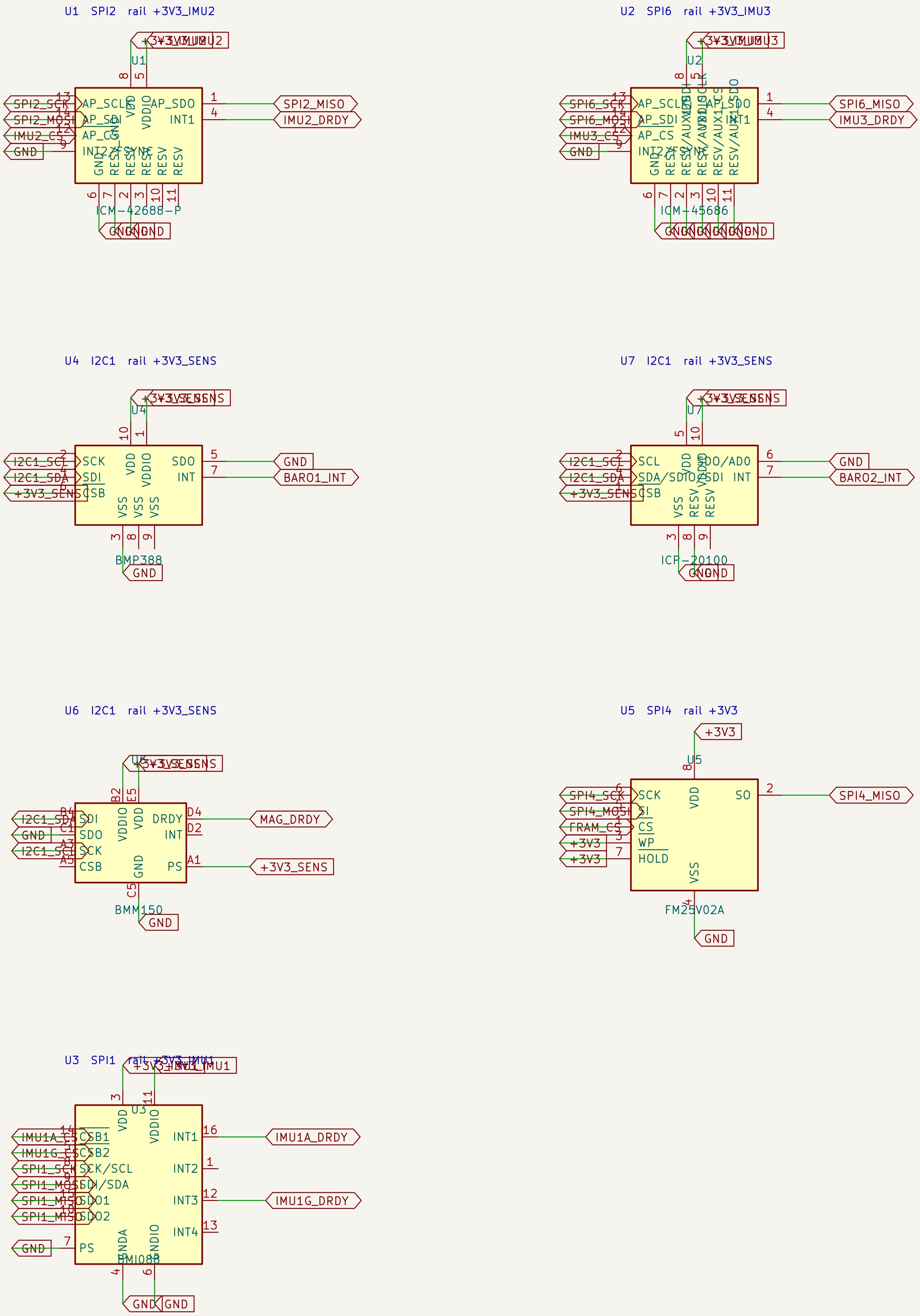
Sheet: /MCU/
File: mcu.kicad_sch

Title: MCU

Size: A4	Date: 2026–08–03	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 2/8

Three IMUs on three separate SPI buses, each on its own switchable rail. One sensor hanging its bus must not take the other two with it.

WARNING — before dropping any +3V3_IMUn rail, firmware must drive that bus's SCK/MOSI/CS low. Interface pins held high with VDDIO off destroy these parts through their ESD diodes (BMP388 datasheet 3.2).



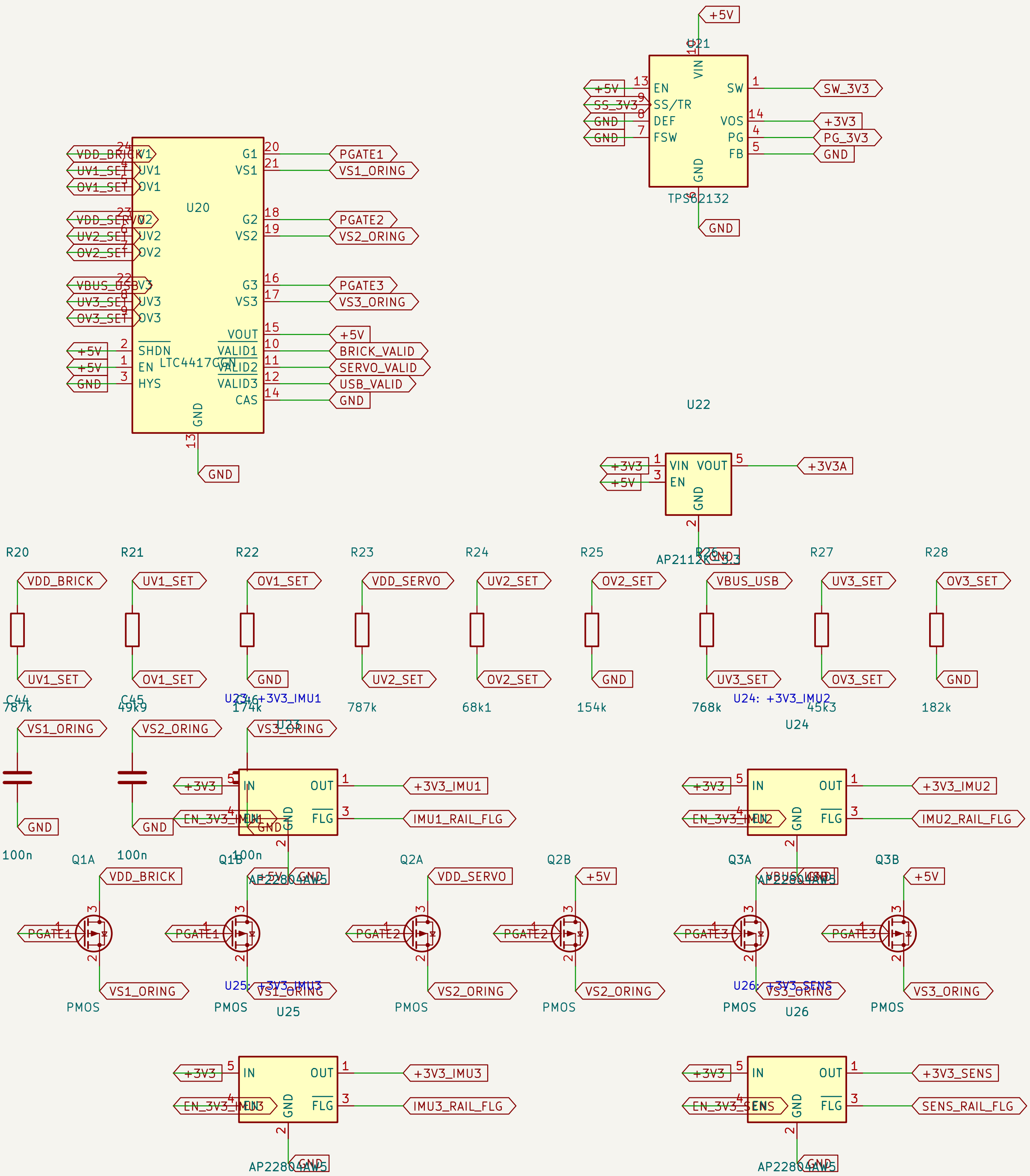
Drive a bus low BEFORE dropping its rail — see ARCHITECTURE.md
Three IMUs, three separate SPI buses, three switchable rails
JFOX Aircraft Co., Ltd.
Sheet: /SENSORS/
File: sensors.kicad_sch
Title: Sensors
Size: A4 | Date: 2026-08-03 | **Rev: A**
KiCad E.D.A. 10.0.5 | Id: 3/8

LTC4417 prioritised ORing: brick > servo rail > USB, with under- and over-voltage lockout per input. This is v2.4.5's arrangement kept, correctly identified – that board's docs called it a BQ24315 until the netlist proved otherwise.

U21 is a BUCK, not an LDO. At 5V in, 3V3 out and the measured ~310 mA typical load an LDO burns 0.53 W, and 0.94 W at peak, which an SOT-23-5 cannot shed – see POWER_BUDGET.md. U22 stays an LDO because it carries only VDDA/VREF+, a few mA, and its quietness is the point.

One load switch per sensor bus, so a wedged IMU can be power-cycled without disturbing the other two.

WARNING – firmware must drive a bus's SCK/MOSI/CS low BEFORE clearing its EN_3V3_* line. Interface pins held high with the rail down destroy these sensors through their ESD diodes (BMP388 datasheet 3.2).



One switchable rail per sensor bus
U21 is a buck (POWER_BUDGET.md); U22 is an LDO for the analog rail
LTC4417 prioritised ORing: brick > servo > USB

JFOX Aircraft Co., Ltd.

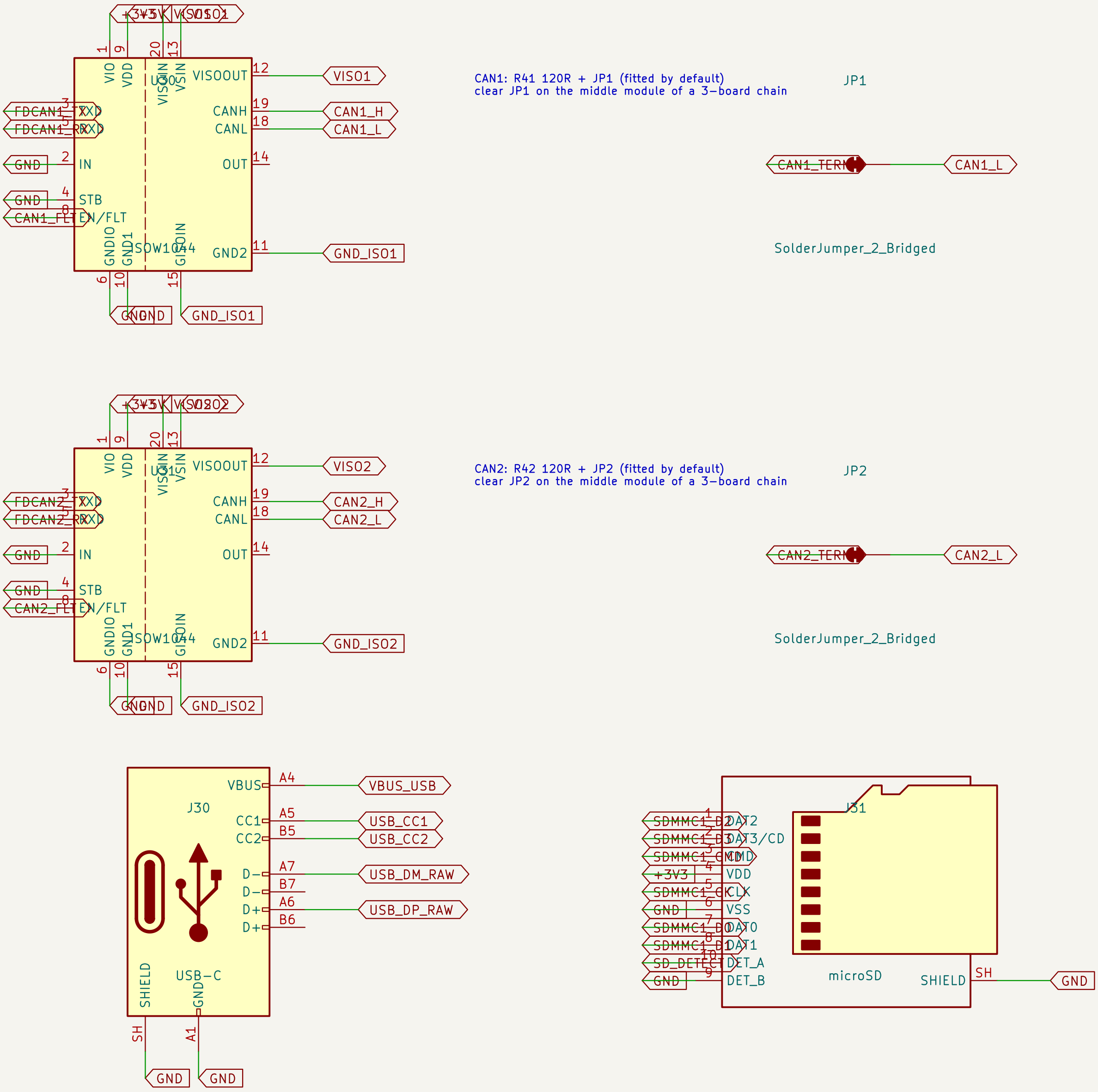
Sheet: /POWER/
File: power.kicad_sch

Title: Power

Size: A4	Date: 2026-08-03	Rev: A
KiCad E.D.A. 10.0.5		Id: 4/8

TWO CAN FD BUSES, BOTH WITH TRANSCEIVERS. v2.4.5 wired CAN2 to the MCU with no transceiver at all, so it was never usable as a bus.

TERMINATION IS SWITCHABLE. 120 ohm in series with a solder jumper, FITTED by default. On a three-board TMR chain the electrically middle module clears JP1 — it does not get R409 desoldered, which is what the old board required. Verify ~60 ohm across CAN_H/CAN_L, bus unpowered, before trusting it.



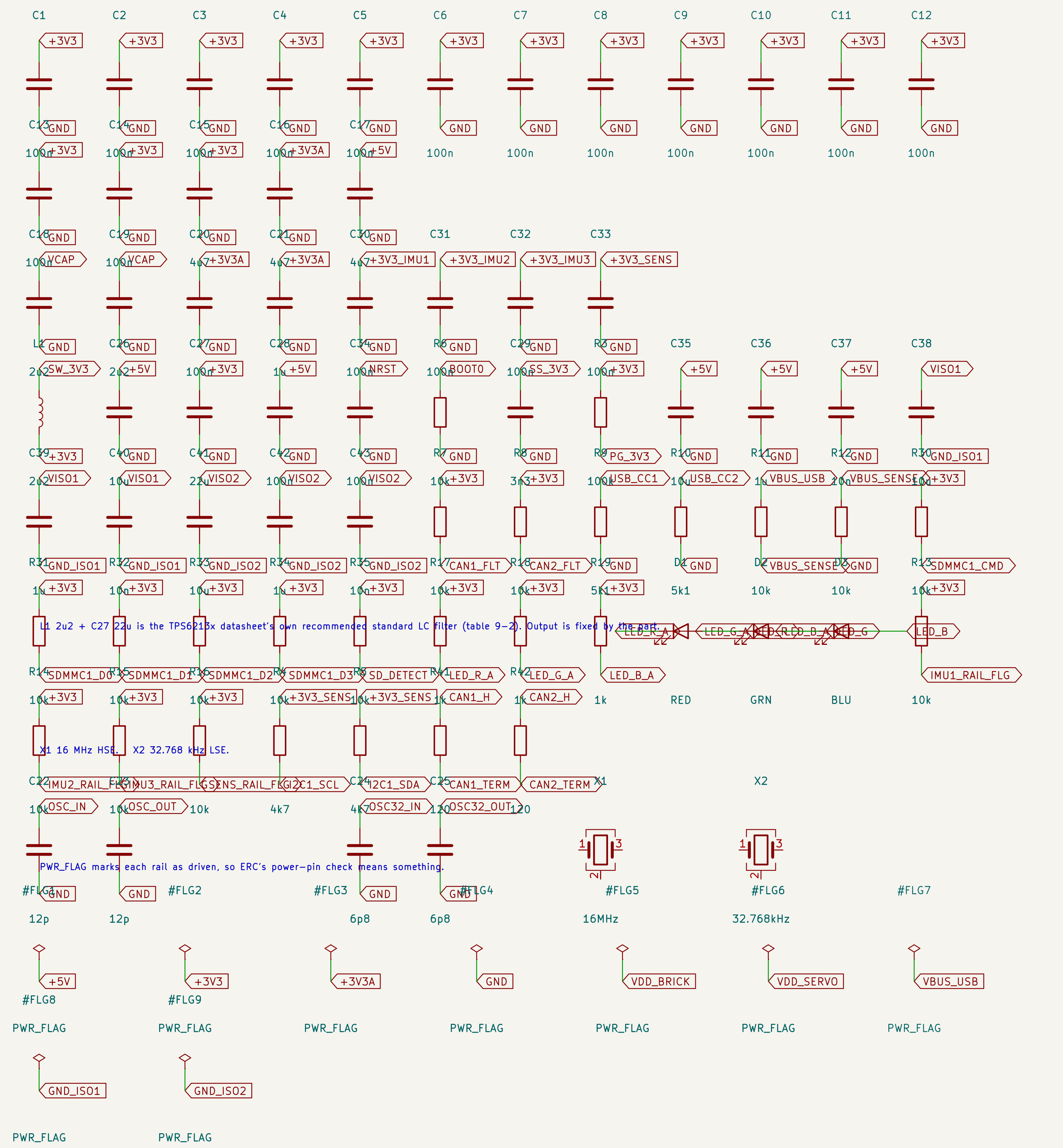
Termination is jumpered: clear JPn on the middle module of a chain
Two CAN FD buses, both with transceivers — v2.4.5 had only one
JFOX Aircraft Co., Ltd.
Sheet: /COMMS/
File: comms.kicad_sch
Title: Comms and Storage

Size: A4	Date: 2026-08-03	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 5/8

Decoupling: one 100n per MCU VDD pin (14), plus 4u7 bulk per rail.
VCAP1/VCAP2 get 2u2 each — the internal LDO's output, not a supply.

16 MHz HSE divides exactly to 48 MHz for USB. The old board's 24 MHz
could not, which left its USB clock at 51.4 MHz.

CAN terminators are HERE, in series with the comms-sheet jumpers.



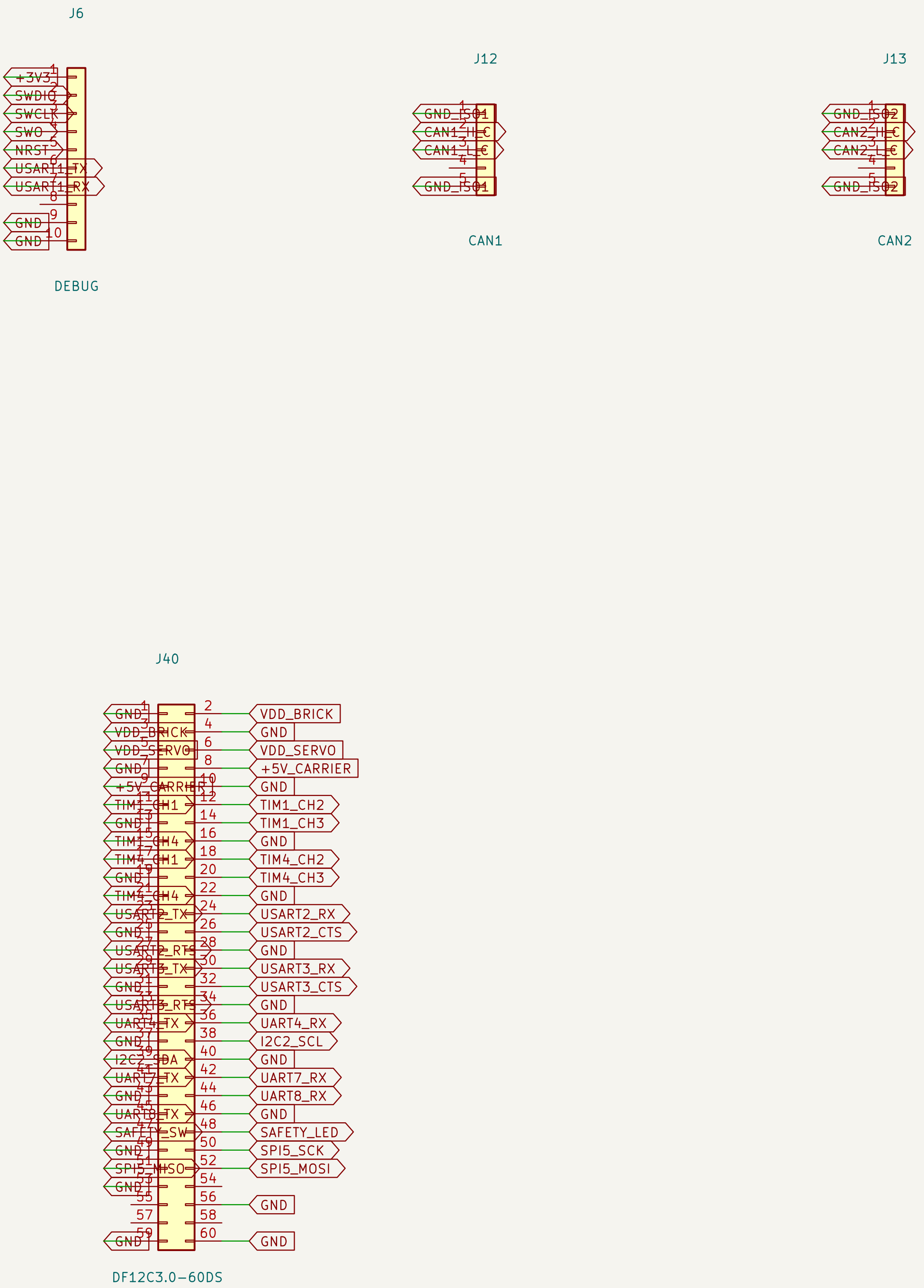
CAN terminators R41/R42 live here, not on the comms sheet
One 100n per MCU VDD pin, 2u2 per VCAP, bulk per rail
JFOX Aircraft Co., Ltd.
Sheet: /PASSIVES/
File: passives.kicad_sch

Title: Passives and Clocks

Size: A4	Date: 2026-08-03	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 6/8

Pinouts follow the Pixhawk connector standard where one exists.
The ecosystem's GPS units, telemetry radios and power modules are
built to it; a novel pinout here means a custom loom for every
peripheral.

JST-GH 1.25 mm throughout – the same family Pixhawk uses, chosen
for its positive latch rather than its size.



TELEM x2, GPS x2, RC, debug, external SPI, 2 power inputs, 8 PWM
Pixhawk-standard pinouts on JST-GH 1.25 mm
JFOX Aircraft Co., Ltd.
Sheet: /CONNECTORS/
File: connectors.kicad_sch

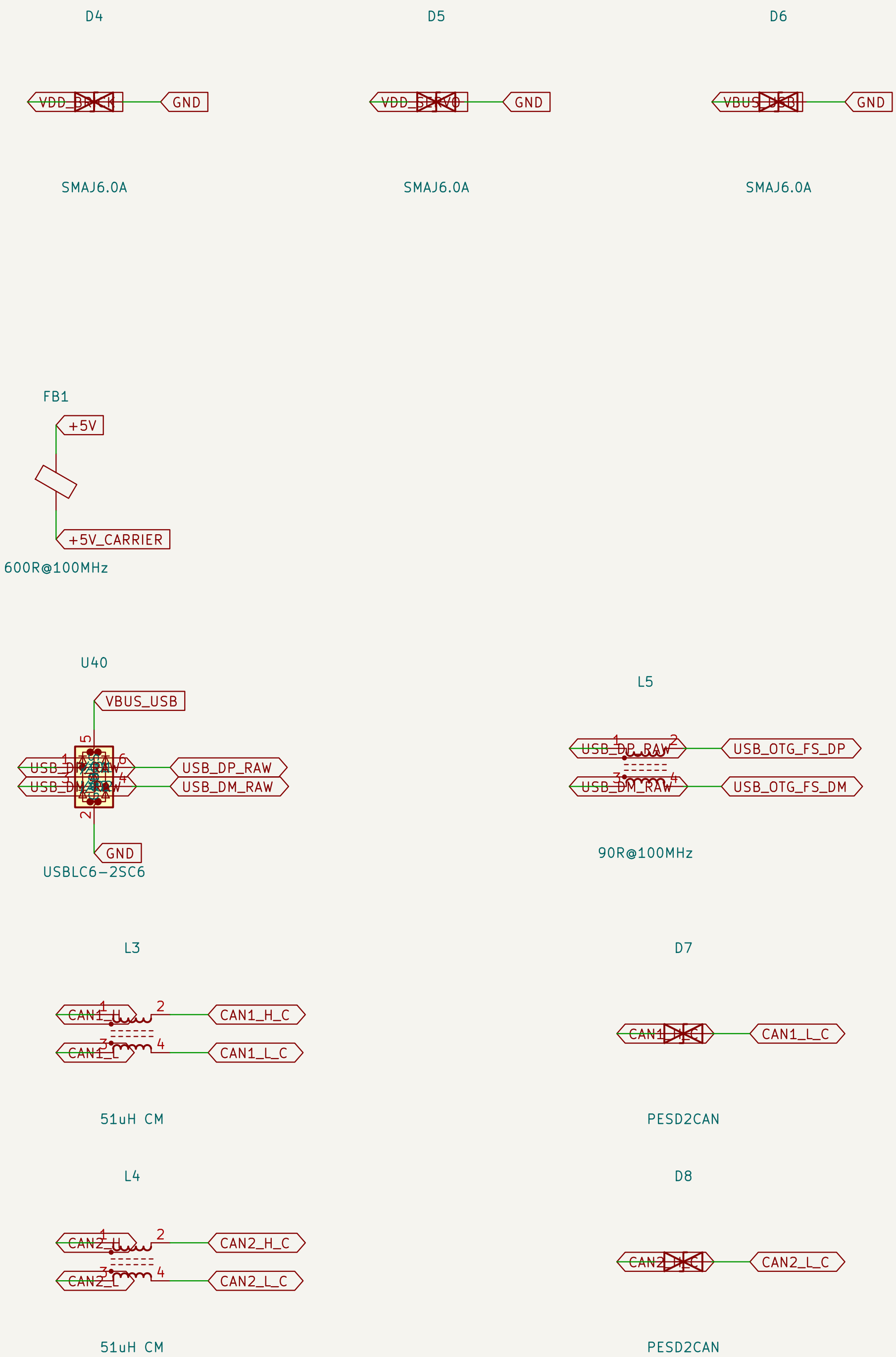
Title: Connectors

Size: A4	Date: 2026-08-03	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 7/8

DO−160G s17 voltage spike, s25 ESD, s21 radiated emissions.

Every off−board harness is an antenna bonded to a board carrying two deliberate switching sources − U21 at 2.5 MHz and each ISOW1044's internal converter at 25 MHz. Suppression at the connector is the only place it works: past it the harness is already radiating.

Placement is the requirement, not the part. Each device belongs AT its connector, ahead of everything it protects, with the shortest return to ground. A TVS 30 mm downstream of the connector protects the last 30 mm of track and nothing else.



CAN protection references GND_ISO, not GND − it is past the barrier

DO−160G s17 voltage spike, s25 ESD, s21 radiated emissions

JFOX Aircraft Co., Ltd.

Sheet: /PROTECTION/

File: protection.kicad_sch

Title: EMI / EMC Protection

Size: A4 | Date: 2026−08−03 | **Rev: A**

KiCad E.D.A. 10.0.5 | Id: 8/8